

A Synopsis
Of
Project
On
CrossLens AI: AI Assistant for Source Review and Consistency
Checking

Submitted in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE & ENGINEERING (AI&ML)

Under the Guidance of
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SEPTEMBER 2026

CERTIFICATE

This is to certify that the project synopsis entitled “**CrossLens AI: AI Assistant for Source Review and Consistency Checking**” is being submitted by Abhinav Chauhan (GE-232023976), Abhilakshya Bhatt (GE-232023975), Deeksha Sharma (GE-232024020), and Aryan Kukreti (GE-232024007), in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering (AI&ML), Department of Computer Science & Engineering, Graphic Era Deemed to be University, Dehradun. The work described in this synopsis is proposed to be carried out under my supervision. The matter embodied in the synopsis has not been submitted to any other university or institution for the award of a degree or diploma.

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DECLARATION

We hereby declare that the project synopsis entitled “CrossLens AI: AI Assistant for Source Review and Consistency Checking”, submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering (AI&ML), Department of Computer Science & Engineering, Graphic Era Deemed to be University, Dehradun, presents our proposed project work. Material drawn from existing publications has been acknowledged through citations and references. This synopsis has not been submitted to any other university or institution for the award of a degree or diploma.

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ABSTRACT

Vehicle insurance claim review requires the comparison of forms, policy clauses, repair tables, and damage photographs. Information distributed across these formats can be difficult to reconcile, particularly when evidence is incomplete or inconsistent.

Recent document-understanding tools, vision-language models, and Retrieval-Augmented Generation (RAG) provide complementary capabilities for extracting structure, interpreting images, and locating relevant knowledge. These capabilities support automation, but retrieved passages and generated descriptions still require verification before they can justify a recommendation.

This project proposes an AI Assistant for Source Review and Consistency Checking, demonstrated through a vehicle insurance claim-review prototype. The system will preserve document structure and source locations, extract normalized claim and estimate records, and obtain structured observations from photographs. It will link repair items to available evidence, retrieve the applicable mock-policy clauses, and use explicit rules for consistency checks and numerical calculations. Findings will distinguish satisfied conditions, contradictions, and insufficient evidence, with uncertain cases referred to a human reviewer. An interactive dashboard will present source-linked explanations and allow corrections. Evaluation will compare the proposed approach with text-only RAG and direct multimodal prompting using extraction quality, finding precision and recall, citation correctness, appropriate deferral, latency, and reviewer effort. The intended contribution is a reproducible verification workflow and evaluation framework built on pretrained models, rather than a new foundation model or a guarantee of autonomous claim settlement.

1 INTRODUCTION

1.1 Background

Enterprise documents communicate information through more than continuous text. A repair estimate conveys meaning through row labels, quantities, prices, and totals; a policy expresses conditions through clauses and amendments; and a photograph records only the visible state of an object. In vehicle insurance review, these sources must be interpreted together to determine which statements agree, which conflict, and which cannot yet be verified.

Document AI has progressed from character recognition toward joint text, image, and layout representations. LayoutLMv3 and Donut illustrate complementary approaches to document understanding [1], [2]. More recent tools and models support structured parsing and visual-language interpretation [6], [8], [9]. Consequently, the relevant challenge is not the complete absence of multimodal capabilities, but their reliable application to a specific review task.

1.2 Motivation

A correctly extracted value is not necessarily a verified fact. A repair total may be arithmetically correct while including an unsupported item; a relevant policy paragraph may omit a referenced exclusion; and a photograph may fail to show the area needed to assess a claimed repair. Treating these situations as equivalent can lead to unsupported recommendations.

The project is motivated by the need to retain the relationships between evidence, checks, and conclusions. A reviewer should be able to inspect the source behind a finding and understand whether the system identified a contradiction or merely lacked sufficient information. Automated processing should reduce repetitive extraction and comparison work while preserving meaningful human oversight.

1.3 Problem Context

The prototype will examine a bounded claim bundle: a claim form, the applicable policy and amendments, an itemized estimate, and vehicle photographs. Repair amounts will initially come from the submitted estimate. The system will check arithmetic and configured deductions; it will not infer a reliable market repair price from photographs alone.

1.4 Scope of the Work

- Document understanding: Preserve text, tables, page references, and bounding boxes from digital and scanned documents.
- Visual evidence: Record visible parts, apparent damage, image identifiers, and visibility limitations.
- Cross-document verification: Compare identifiers, dates, estimate items, photographic observations, and applicable mock-policy conditions.
- Decision support: Present traceable findings and retain unknown results when evidence is unavailable.
- Evaluation: Use separate development and test bundles; label synthetic material and prevent overlap between splits.

Autonomous settlement, definitive fraud detection, hidden-damage diagnosis, and foundation-model training are outside scope. Other enterprise domains remain future extensions.

1.5 Significance and Proposed Contribution

The proposed contribution is an integrated evidence representation and verification workflow. Each finding will connect the relevant extracted facts, repair item, policy condition, and source location. This organization will support reproducible checks and correction of upstream extraction errors without treating a fluent model response as proof.

Evaluation will test whether evidence linking and coded checks reduce unsupported findings. RAG supplies policy context; optional MCP interfaces expose backend functions. Adopting these technologies alone does not establish research novelty.

2 LITERATURE REVIEW

The following ten studies cover document understanding, retrieval, and vehicle damage. Their contributions and limitations inform the proposed claim-review workflow; the comparison table and research gap distinguish existing capabilities from the work to be evaluated.

2.1 LayoutLMv3

Huang et al. [1] jointly model text, image patches, and layout through unified masking and word-patch alignment. Their results motivate retaining spatial relationships in forms and estimates. However, task-level document understanding does not verify an entire claim across files; the proposed application must separately associate extracted facts and validate their consistency.

2.2 Donut

Kim et al. [2] introduce Donut, a Transformer that interprets document images without a separate OCR stage. This reduces dependence on a conventional recognition pipeline, but generated fields still require validation. The project will select extraction tools using field accuracy, table recovery, and source traceability on representative claim documents.

2.3 LLaVA

Liu et al. [3] connect a vision encoder and language model through visual instruction tuning. LLaVA demonstrates image-based conversation and motivates structured descriptions of damage photographs. Such descriptions cannot establish hidden damage, accident causation, or replacement necessity; visible observations and unresolved evidence must therefore remain separate in the proposed system.

2.4 Retrieval Augmented Generation

Lewis et al. [4] combine a generator with retrieved external passages for knowledge-intensive tasks. RAG provides the foundation for locating relevant policy clauses, but relevance does not establish applicability or completeness. The prototype must filter by policy version, include amendments, and apply reviewed rules independently of the generated explanation.

2.5 ColPali

Faysse et al. [5] introduce ColPali, using page-image representations and late interaction to retrieve visually rich documents. It can retain cues lost during text extraction. The project will consider visual retrieval when text retrieval misses important policy content; matching a retrieved page to a repair item still requires separate verification.

2.6 Docling

Livathinos et al. [6] present Docling, a modular toolkit for conversion into a unified document representation with layout and table structure. It provides a practical ingestion foundation for claim forms and estimates. Extracted values must still be validated, with page locations retained so a reviewer can inspect and correct the original evidence.

2.7 OmniDocBench

Ouyang et al. [7] introduce OmniDocBench for diverse document-parsing evaluation, including component and attribute-level analysis. Its approach supports separate assessment of tables, reading order, and difficult scans. General parsing scores cannot establish insurance-review accuracy, so this project also needs annotated claim bundles and evidence-supported finding labels.

2.8 Qwen3 VL

Bai et al. [8] describe Qwen3-VL, a contemporary family of vision-language models for interleaved multimodal inputs. It is a candidate for structured photographic observations and explanations. Selection will depend on local evidence accuracy, unsupported observations, latency, and memory use; general benchmarks do not establish reliability on insurance claims.

2.9 PaddleOCR VL 1.6

Zhang et al. [9] propose PaddleOCR-VL-1.6, using region refinement and progressive post-training to improve document parsing. The reported 96.33 score on OmniDocBench v1.6 concerns that benchmark, not this project. Adoption will depend on measured improvements over the initial parser on local scans and estimates, together with deployment cost.

2.10 CarDD

Wang, Li, and Wu [10] introduce CarDD, containing 4,000 damage images and over 9,000 annotated instances across six categories. It supports visual detection and segmentation evaluation, but contains no matched policy or settlement records. The team must construct separate claim bundles, label simulated associations, and prevent related images from crossing evaluation splits.

2.11 Comparative Analysis

Table 2.1 compares the role of each reviewed study and the additional work needed for the proposed claim-review application.

Study	Main contribution	Implication for this project
[1] LayoutLMv3	Joint text, image and layout learning	Preserve structural associations; add cross-file verification.
[2] Donut	Document interpretation without separate OCR	Evaluate field reliability and source traceability.
[3] LLaVA	Visual instruction following	Validate damage observations and visibility limits.
[4] RAG	Generation using retrieved knowledge	Filter policy versions and apply explicit checks.
[5] ColPali	Retrieval from page-image representations	Evaluate need beyond text retrieval; link repair evidence separately.
[6] Docling	Unified structured conversion	Validate extracted values and retain original evidence.
[7] OmniDocBench	Diverse parsing evaluation	Add annotated claim-level verification cases.
[8] Qwen3-VL	General multimodal perception and generation	Benchmark insurance evidence accuracy and resource use.
[9] PaddleOCR-VL-1.6	Refinement of difficult parsing regions	Measure benefits on local scans and estimates.
[10] CarDD	Annotated vehicle-damage imagery	Construct separate policies, estimates and review labels.

The comparison identifies implementation and evaluation needs for this prototype. It does not establish that the combined approach is globally novel or that any listed model is universally superior.

2.12 Current State of the Art and Research Gap

Current research offers complementary capabilities: structured parsers for ingestion, VLMs for interpretation, and retrieval models for evidence selection. No single benchmark score establishes performance across the full review task. The project will investigate how these components can support a reproducible verification procedure, using annotated insurance cases and explicit uncertainty handling. Its contribution must be established experimentally rather than assumed from integration alone.

- Evidence association: The application must connect a repair line item to the correct vehicle part, available photographs, and relevant policy condition while retaining the evidence behind each association.

- Uncertainty representation: A contradiction, an unreadable value, and an unphotographed vehicle part require different reviewer actions. Missing evidence must not automatically be interpreted as disproof.
- Policy applicability: Similar wording is insufficient if it comes from the wrong policy version or omits an amendment. Mandatory checks need an explicit reviewed checklist.
- Reproducible verification: Arithmetic, identifier comparisons, and configured policy conditions should produce repeatable results with rule versions and source references.
- Evaluation of added value: Component accuracy must be distinguished from overall review quality. Baselines and ablations should establish whether evidence linking and coded checks improve results.

2.13 Research Question

Does explicit cross-document evidence linking, combined with rule verification and appropriate deferral, reduce unsupported findings in multimodal claim review compared with text-only RAG and direct multimodal prompting?

The comparison will consider finding precision and recall, citation support, and reviewer effort at comparable levels of answered and deferred cases. Proposed improvements will be reported as findings only after evaluation; no performance gain is assumed in advance.

3 PROBLEM STATEMENT AND OBJECTIVES

3.1 Problem Statement

Vehicle insurance claim bundles distribute relevant information across forms, policy clauses, repair tables, and photographs. Although existing models can extract and interpret these inputs, reliable review requires their outputs to be connected, checked, and traced to evidence. Incorrect extraction, mismatched items, incomplete policy context, and limited photographic visibility can otherwise produce unsupported findings.

The problem addressed is how to build and evaluate a bounded multimodal verification workflow that identifies inconsistencies, applies reviewed mock-policy conditions, and distinguishes insufficient evidence from failed conditions while supporting human review.

3.2 Objectives

- To extract text, table values, document structure, and source locations from heterogeneous claim documents.
- To obtain structured visual observations while recording the limits of the submitted photographs.
- To normalize and link claim facts, repair items, photographs, and applicable policy clauses.
- To retrieve relevant policy context and execute reviewed rules for consistency checks and numerical calculations.
- To present evidence-backed findings, uncertainty, and correction options through an interactive dashboard.
- To construct annotated evaluation cases and compare the workflow with text-only RAG and direct multimodal prompting, including ablation studies of evidence linking and rule checks.

4 PROPOSED METHODOLOGY

The project will implement a staged decision-support workflow for vehicle insurance claim bundles. Pretrained document and vision-language models will supply structured evidence, while retrieval and reviewed rules will support traceable checks. Human reviewers will resolve uncertain cases and retain responsibility for final decisions.

4.1 High Level Architecture

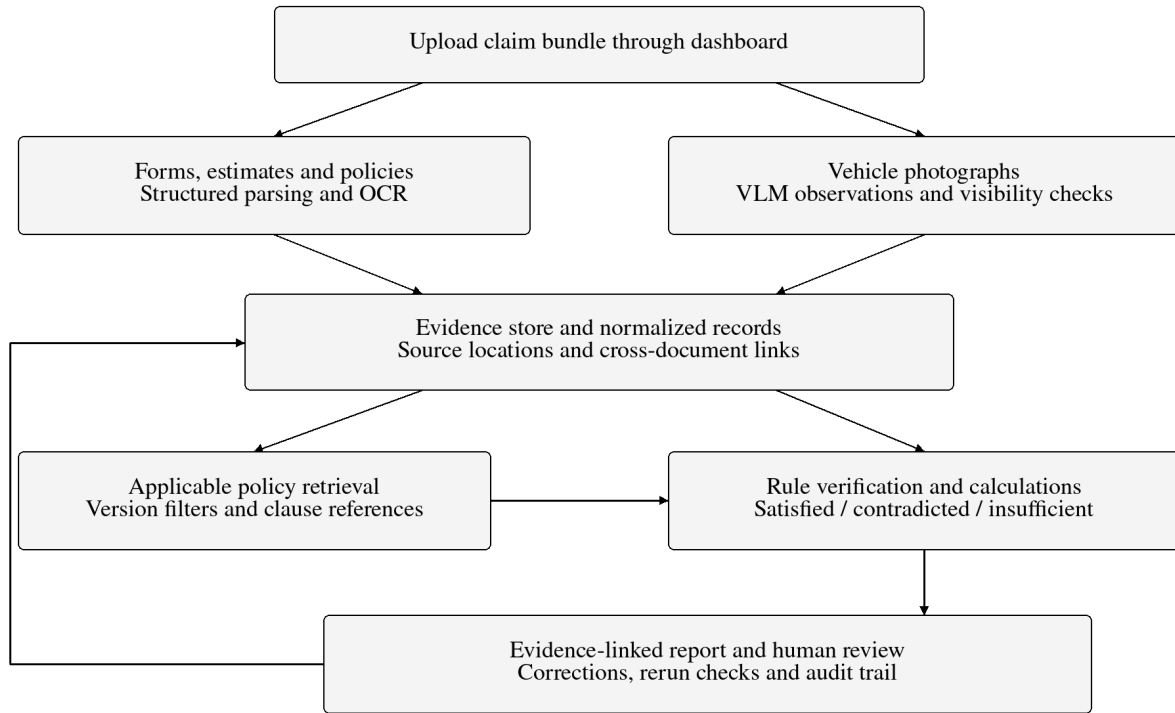


Figure 4.1 Proposed architecture for evidence-based claim review

The evidence store will retain original files, page locations, extracted records, and their links. Retrieval will select applicable policy context; the verification service will produce individual findings with source references. Optional MCP adapters may expose these backend functions through a standard tool interface.

4.2 Processing and Verification Workflow

1. Ingest and structure the bundle. Validate uploaded file types, assign claim and document identifiers, and preserve originals. Use Docling or PyMuPDF for document extraction, with OCR where needed. Normalize form fields and estimate rows into Pydantic schemas while retaining page numbers, bounding boxes, and extraction warnings.
2. Observe and link visual evidence. Use an evaluated VLM, with Qwen3-VL as a candidate, to record visible parts, apparent damage, and visibility limits. Link observations to estimate items through normalized part names and evidence references. An unseen part will be marked as insufficient evidence rather than automatically treated as undamaged.
3. Retrieve applicable policy clauses. Use hybrid full-text and vector retrieval over versioned policy sections, followed by reranking where useful. Restrict retrieval to the relevant mock policy and amendments, and include definitions needed for each check. A reviewed checklist will identify mandatory conditions that retrieval alone could overlook.
4. Verify facts and amounts. Apply versioned Python rules to identifiers, dates, evidence links, and configured policy conditions. Read repair amounts from submitted estimate rows; recompute quantity $\times$ unit rate and totals, then apply only reviewed deductions and limits. The prototype will validate submitted costs, rather than infer market prices from photographs. Each check will return satisfied, contradicted, or insufficient evidence with its supporting sources.
5. Report, review, and evaluate. Present findings, calculations, and evidence in a dashboard; allow corrections and rerun affected checks with an audit trail. On held-out annotated bundles, compare text-only RAG, direct multimodal prompting, and the proposed workflow. Measure extraction quality, finding precision and recall, citation correctness, deferral, latency, and reviewer effort. Ablations will assess the contribution of evidence linking and coded rules.

5 HARDWARE AND SOFTWARE REQUIREMENTS

The following resources support development and prototype evaluation. Final deployment choices will depend on the selected models, document volume, and measured resource requirements.

5.1 Hardware Requirements

- Development workstation: 16 GB RAM is a practical starting configuration for the interface, database, and lighter processing stages; higher memory is preferable for concurrent services.
- Model inference: Use a separate local or cloud GPU service. Select GPU memory after profiling the chosen model, quantization, image resolution, context length, and batch size. An 8 GB GPU is not a universal minimum for all configurations.
- Storage: Reserve SSD capacity for documents, page renders, model weights, and test datasets. A 250 GB SSD can be a starting allocation; actual space depends on the selected models and dataset size.

5.2 Software Requirements

Layer	Proposed technology
Language and validation	Python; Pydantic schemas
Document processing	Docling; PyMuPDF; OCR where needed
Models and inference	PyTorch; Hugging Face Transformers; evaluated VLM
Retrieval and storage	PostgreSQL; pgvector; full-text search; reranking
Verification	Versioned Python rules and reviewed mock-policy configuration
Backend and workflow	FastAPI; explicit workflow or LangGraph
User interface	React with a document viewer; Streamlit for an early prototype
Testing and integration	pytest; Git/GitHub; Docker; optional MCP interface

Final model and package versions will be recorded with evaluation results. Existing pretrained components will be reused; the project does not require training a foundation model.

6 EXPECTED OUTCOMES

The project is expected to deliver a working prototype and an evaluation of its usefulness for evidence-based claim review. The following outcomes describe intended deliverables; accuracy improvements and time savings will be established through testing rather than assumed in advance.

- A functioning multimodal workflow that accepts a claim form, policy, itemized estimate, and photographs, and produces normalized records with links to the original evidence.
- A structured review report identifying supported conditions, contradictions, and insufficient evidence, together with the policy clauses and document locations behind each finding.
- Reproducible numerical checks that recalculate estimate line items and totals and demonstrate configured deductions or limits using reviewed mock-policy rules.
- An interactive dashboard through which a reviewer can inspect evidence, correct extracted facts, request missing information, and review updated findings with an audit trail.
- An annotated evaluation set with documented development and test splits, including clearly labeled synthetic cases, missing evidence, and cross-document inconsistencies.
- A comparative evaluation against text-only RAG and direct multimodal prompting, reporting extraction quality, finding precision and recall, citation correctness, appropriate deferral, latency, and reviewer effort.

Together, these outputs should demonstrate whether explicit evidence links and rule checks improve the reliability of a bounded review workflow. The prototype will support human decisions; it will not perform autonomous claim settlement.

7 CONCLUSION

This synopsis proposes a practical approach to reviewing vehicle insurance documents and photographs as a connected evidence bundle. The central objective is to make automated findings traceable, reproducible, and open to correction by a human reviewer.

The system will combine structured document extraction, visual observations, policy retrieval, and explicit verification rules. Preserving source locations and distinguishing contradictions from insufficient evidence should make the basis of each recommendation easier to inspect. Repair calculations will use submitted estimate values and reviewed mock-policy conditions, with uncertainty retained where the available material cannot support a conclusion.

The project contribution will be assessed through comparisons with text-only RAG and direct multimodal prompting, supported by component ablations and annotated test cases. Results will establish the benefits and limitations of the proposed integration. Extension to other enterprise workflows will remain future work after the insurance prototype has been evaluated.

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